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CENTRE
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CHEMISTRY

Grade 12

Paper 2

May 2022

2 hours

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Ruler
 Data Booklet

12CHEM/02

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may receive fewer marks if you do not show your working or do not use appropriate units where required.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 100.

Answer all questions in English.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

This document consists of **18** printed and **2** blank pages.

1 This question is about isotopes and ionisation energy.

(a) Most atoms have several naturally occurring isotopes.

(i) In terms of sub-atomic particles, state the difference between isotopes of the same element.

.....
..... [1]

(ii) An alloy contains aluminium and copper. The table gives the mass number of isotope in the alloy.

mass number of isotope	27	63	65
relative abundance	20	124	56

Write the symbol of the isotope with mass number of 27.

..... [1]

(iii) Use the appropriate relative abundances to calculate the relative atomic mass of copper. Give your answer to **one** decimal place.

Show your working.

[2]

(b) The atoms in the alloy must be ionised before it can be analysed in a mass spectrometer.

Write an equation, including state symbols, that represents the first ionisation energy of aluminium.

..... [1]

(c) State **one** reason why the first ionisation energy of aluminium is lower than the first ionisation energy of magnesium.

.....
.....
..... [1]

[Total: 6]

- 2 Most of the metal cations of the transition elements exist as the hexa-aqua complex in aqueous solution.

For
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- (a) Explain why aqueous copper(II) ions are coloured.

.....

.....

.....

.....

..... [3]

- (b) Diaminoethane can react as a base **and** a ligand in the same way as ammonia.

- (i) Write an equation for the complete ligand exchange reaction between aqueous copper(II) ions and 1,2-diaminoethane.

..... [2]

- (ii) Explain why the entropy change for this reaction is positive.

.....

..... [1]

- (c) Write an equation for the reaction between aqueous copper(II) ions and sodium carbonate.

..... [2]

[Total: 8]

3 This question is about the halogens and compounds containing halogens.

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(a) The electronegativities of the halogens are shown in the table.

halogen	fluorine	chlorine	bromine	iodine
electronegativity	4.0	3.5	2.8	2.5

(i) Define the term *electronegativity*.

.....

 [1]

(ii) Explain the trend in electronegativities shown by the halogens.

.....

 [2]

(b) An aqueous solution of iodide ions reacts with aqueous bromine water.

Write the ionic equation for this reaction.

..... [1]

(c) The boiling point of hydrogen chloride, HCl is -85°C and of iodine(I) chloride, ICl is 97°C

Explain, in terms of intermolecular forces, the difference in boiling point between HCl and ICl .

.....

 [2]

(d) Halogens react with metals to form ionic compounds.

Use the following data and Data booklet to calculate the lattice energy, ΔH_{latt} , of calcium bromide, $\text{CaBr}_2(\text{s})$.

You may find it helpful to construct an energy cycle.

type of enthalpy change	value / kJ mol^{-1}
enthalpy change of formation CaBr_2	−675
1st ionization energy of calcium	+ 590
2 nd ionization energy of calcium	+ 1145
1 st electron affinity of bromine	−325
enthalpy change of atomisation of calcium	+ 178
enthalpy change of vaporization of bromine	+ 31

ΔH_{latt} kJ mol^{-1} [3]

[Total: 9]

4 This question is about acids and bases.

(a) Define the term *Brønsted-Lowry base*.

.....
..... [1]

(b) Methylamine, CH_3NH_2 , is a Brønsted–Lowry base.

(i) Write an equation for the reaction of methylamine with water to form a weakly alkaline solution.

..... [1]

(ii) State why the solution formed is weakly alkaline.

.....
..... [1]

(iii) For the reaction in (b)(i), give the formula of the Brønsted-Lowry base and of its conjugate acid.

conjugate base [1]

(c) An excess of methylamine reacts with hydrochloric acid to form an alkaline buffer solution.

Using this solution as an example explain why the buffer solution resists the change in pH when a small amount of sodium hydroxide is added

.....
.....
..... [2]

- (d) The pH of methanoic acid solution is 2.38. The value of K_a for methanoic acid is $1.60 \times 10^{-4} \text{ mol dm}^{-3}$ at 25°C .

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Calculate the concentration of methanoic acid. Show your working.

concentration of methanoic acid $\text{kJ mol}^{-1} \text{ dm}^{-3}$ [3]

- (e) 1 dm^3 of buffer solution is made from methanoic acid with the concentration determined in part 4(d) and $0.150 \text{ mol dm}^{-3}$ sodium methanoate.

A sample of sodium hydroxide containing 0.0150 mol of NaOH is added to this buffer solution.

Calculate the pH of the buffer solution after the sodium hydroxide has been added. Give your answer to **two** decimal place.

pH is [5]

[Total: 14]

5 Alkenes are unsaturated hydrocarbons.

For
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(a) Explain the meaning of the term *unsaturated*.

.....
..... [1]

(b) Unsaturated hydrocarbons can form several types of isomers.

Draw **and** name **two** stereoisomers of pentene.

[2]

(c) Pent-1-ene reacts with hydrogen bromide at room temperature and pressure.

(i) Write an equation for this reaction and name the major product.

equation
name [2]

(ii) Name **and** draw the mechanism of the reaction using 'curly arrows'.

equation

mechanism

[5]

- (d) (i) Alkenes produce many useful polymers. Polychloroethene is made from $\text{CH}_2=\text{CHCl}$.

Draw the structure of **two** repeating units of polychloroethene.

For
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[1]

- (ii) Polychloroethene is a non-biodegradable polymer.

Explain the meaning of the term *non-biodegradable*.

.....

..... [1]

[Total: 12]

6 This question is about rate of reaction and chemical equilibria.

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(a) The activation energy, E_a , for a reaction involving the gas is shown in Fig. 6.1.

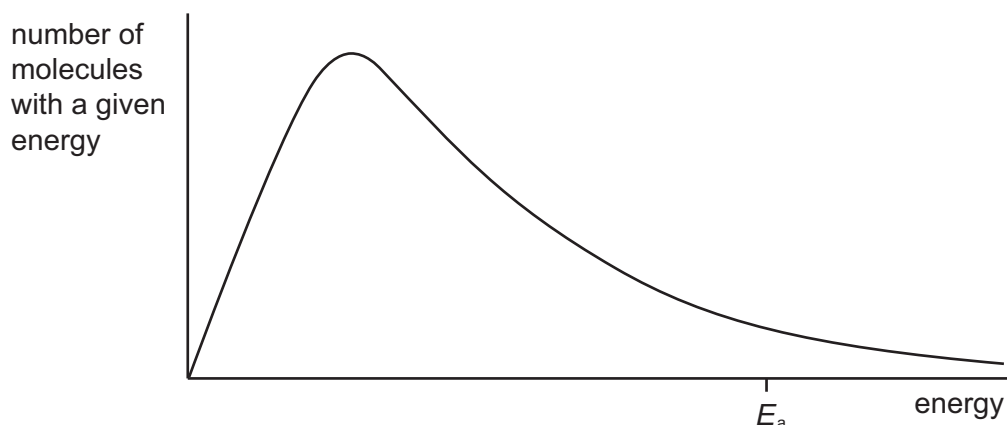


Fig. 6.1. Maxwell-Boltzmann distribution

Label on Fig. 6.1 the activation energy, E_{cat} , when a catalyst is present.

Sketch on Fig. 6.1 the Maxwell-Boltzmann distribution curve for the same sample of gas at a **lower** temperature. Label this curve T_L .

[2]

(b) Two gases, **A** and **B**, react to form compound **C**.



The initial rate of this reaction is measured in a series of experiments at a constant temperature.

The rate equation is determined.

$$\text{rate} = k [\mathbf{A}]^2[\mathbf{B}]$$

experiment	initial [A] / mol dm ⁻³	initial [B] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.20	0.30	1.8×10^{-3}
2	0.40	0.60	to be calculated

- (i) Use the data from experiment 1 to calculate a value for the rate constant k at this temperature **and** deduce its units. Give your answer to **two** decimal place.

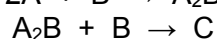
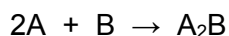
rate constant

unit [2]

- (ii) Calculate a value for the initial rate in experiment 2. Quote your answer to **two** significant figures.

[1]

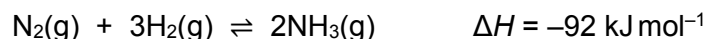
- (iii) A proposed **two-step** mechanism for the reaction $A + B \rightarrow C$ is the following:



Identify a rate determining step using the rate equation.
Explain your answer.

..... [1]

- (c) A sample of nitrogen, hydrogen and ammonia in a closed container is allowed to reach equilibrium at a constant temperature.



- (i) Give **two** features of dynamic equilibrium.

feature 1

.....

feature 2

..... [2]

- (ii) The volume of the container is increased at the same temperature. No substances are added or taken away from the container.

State the effect, if any, on the equilibrium amount of $N_2(g)$ within the larger container.

..... [1]

- (iii) The temperature **and** pressure of the equilibrium mixture is increased.

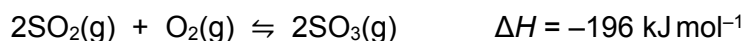
State the effect on equilibrium constant, K_c .

Explain your answer.

.....

..... [2]

- (d) (i) The dynamic equilibrium shown is established at temperature, T_1 , in a closed container.



The value of K_c for the reaction is $68.0 \text{ dm}^3 \text{ mol}^{-1}$ when the equilibrium mixture contains 3.82 mol of O_2 and 5.24 mol of SO_3 .

The volume of the container is 10.0 dm^3 .

Calculate the concentration, in mol dm^{-3} , of SO_2 in the equilibrium mixture. Quote your answer to **three** significant figures.

concentration of SO_2 mol dm^{-3} [3]

- (ii) The temperature of the equilibrium mixture in (d) is changed to T_2 and the mixture is left to reach a new equilibrium position. At this new temperature the equilibrium concentration of SO_3 has increased.

State **and** explain what conclusion you can make about the new temperature T_2 .

.....

.....

.....

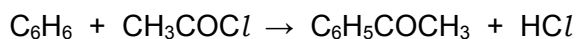
.....

..... [3]

[Total: 17]

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- 7 Benzene reacts with ethanoyl chloride in the presence of an aluminium chloride catalyst. This reaction is an example of electrophilic substitution.



- (a) Ethanoyl chloride reacts with the catalyst aluminium chloride, AlCl_3 , to form an electrophile and anion **Z**.

- (i) Write the formulas of electrophile **and** anion **Z**.

electrophile

anion **Z** [2]

- (ii) Draw a 'dot and cross' diagram **and** state the bond angle in anion **Z**.

Show outer shell electrons only.

[2]

- (iii) Draw the mechanism for the reaction between benzene and the electrophile in (a)(i) using 'curly arrows'.

[3]

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- (b) A racemic mixture (racemate) of compound **A** is formed in the two-stage synthesis shown in Fig 7.1.

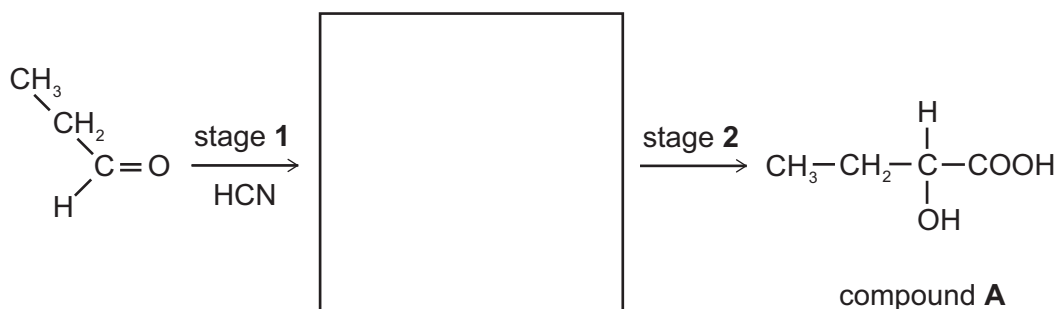


Fig. 7.1. Forming compound **A**

- (i) Draw a structural formula of the compound in the box. [1]
- (ii) Give the IUPAC name for compound **A**. [1]
-
- (iii) State the type of mechanism for the reaction in stage 1. [1]
-
- (c) (i) Give the meaning of the term *racemic mixture* (racemate). [1]
-
-
- (ii) Draw optical isomers of compound **A**. [2]
-
-
- (iii) Optical isomers are used in drug production in pharmacology. [2]
- Suggest **two** reasons why pure enantiomers is better option for use as pharmaceutical drugs rather than racemic mixtures
-
-
-
-

[Total: 15]

- 8 This question is about amino acids and their properties. Use the Data Booklet to answer to the following questions.

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(a) (i) Draw the structure of the zwitterion formed by cysteine.

[1]

(ii) Draw the structure of the species formed by lysine at pH 2.

[1]

(b) Two aspartic acid molecules and one cysteine molecules react to form a tripeptide with the order aspartic acid-cysteine-aspartic acid.

Draw the structure of this tripeptide.

[2]

(c) (i) Aspartic acid reacts with excess aqueous sodium hydroxide.

Write an equation for this reaction.

[2]

- (ii) Aspartic acid reacts with excess hydrochloric acid.

Write an equation for this reaction.

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- (d) (i) State the type of linkage between cysteine residues.

[2]

.....

..... [1]

- (ii) Proteins have several types of structures.

State the type of the structure that is stabilised by the linkage in **d(i)**.

..... [1]

[Total: 10]

- 9 Compound **G**, $\text{C}_4\text{H}_6\text{O}_2$, is analysed by infrared and proton NMR spectroscopy. The spectrums of **G** are shown below.

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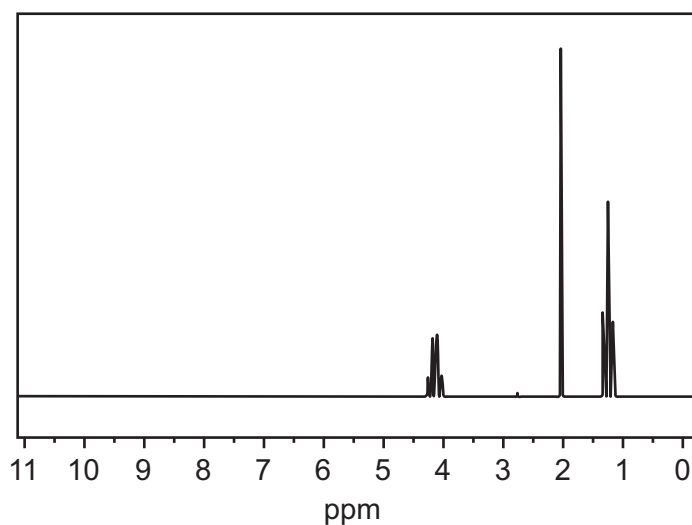


Fig. 9.1. The proton NMR spectrum of compound **G**

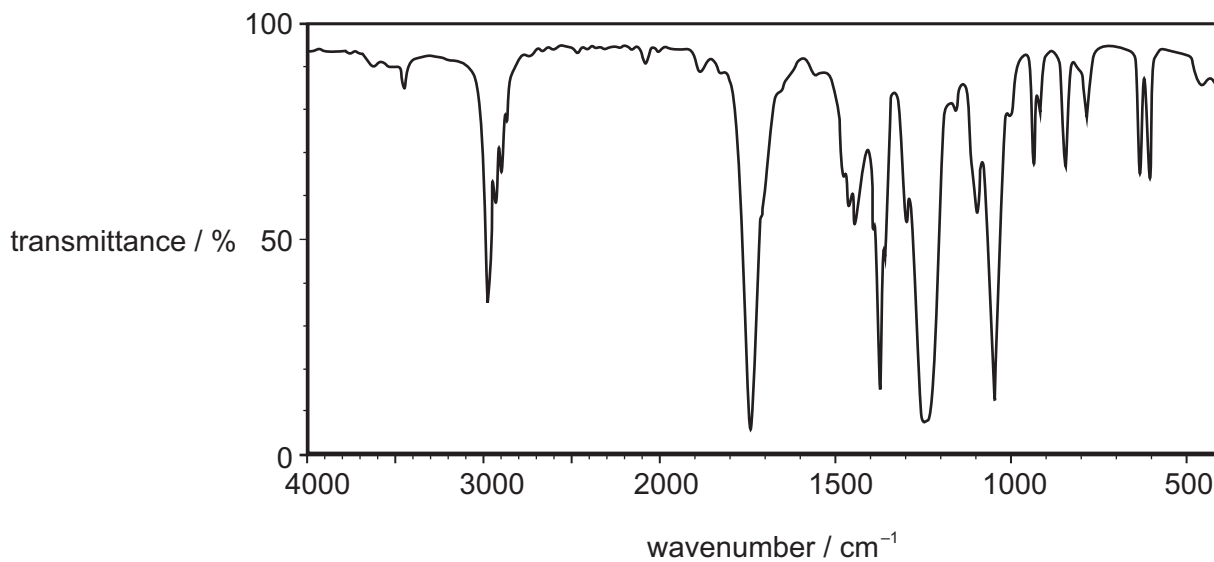


Fig. 9.2. The IR spectrum of compound **G**.

- (a) (i) Identify a compound used as the reference standard in proton NMR spectroscopy.

..... [1]

- (ii) Explain the position and splitting of the following peaks and absorption.

quartet peak at $\delta = 4.1$ ppm

.....

triplet peak at $\delta = 1.2$ ppm

.....

1720 cm^{-1} in the IR spectrum

..... [3]

- (b) Write the structural formula of compound **G**.

[1]

- (c) Compound **G** has several isomers.

Draw **and** name **two** isomers of compound **G** that contain the same functional group as **G**.

name 1

name 2 [4]

[Total: 9]

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